

Campylobacter

**Background document for the
WHO *Guidelines for drinking-water quality* and the
WHO *Guidelines on sanitation and health***



**World Health
Organization**

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Highlights

- There is strong evidence linking transmission of *Campylobacter* with unsafe drinking-water, inadequate sanitation, and poor hygiene. *Campylobacter* are major zoonotic waterborne pathogens. Multiple drinking-water outbreaks of campylobacteriosis have been documented, particularly in small groundwater systems.
- Poultry, cattle, sheep and other domestic animals, wild fowl, pets and humans are important sources of *Campylobacter*.
- Transmission is primarily by the faecal–oral route. Food is the most common source of infection; consumption of contaminated water and direct contact with animals are also significant sources of exposure. Person-to-person transmission is uncommon.
- Infection may be asymptomatic or cause mild to severe illness. Symptoms include abdominal pain, acute watery diarrhoea, vomiting, chills and fever. *Campylobacter* can cause serious post-infection disease, including reactive arthritis, Guillain-Barré syndrome and possibly inflammatory bowel disease and irritable bowel syndrome.
- Campylobacteriosis is generally self-limiting but in severe cases treatment with antibiotics may be beneficial, but strains of *Campylobacter* with resistance to multiple antibiotics have been reported.
- In low-income settings, *Campylobacter* infection is often endemic, with most cases in children under 2 years of age. In high-income settings, the age distribution is wider, but the burden is highest in children under 5 years of age.
- The relative infectivity of *Campylobacter* is considered moderate to high. Estimates of the ID₅₀ (50% infective dose) for *C. jejuni* from challenge studies and analysis of outbreaks vary from less than 10 to 890 organisms.
- Culture-based and molecular methods for *Campylobacter* detection in environmental samples are available.
- *Campylobacter* are commonly present in sewage, treated wastewater and surface water, and have been detected in vulnerable groundwater and inadequately treated drinking-water supplies. The persistence of *Campylobacter* in water is considered to be short.
- Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains, are needed to manage microbial risks, including from *Campylobacter*. Since animals are an important source of *Campylobacter*, a comprehensive approach that also deals with the management of animal faeces is necessary to control transmission.
- Reduction of *Campylobacter* by wastewater and drinking-water treatment varies depending on the process but is similar to the reduction of faecal indicator bacteria such as *E. coli*. *Campylobacter* have low resistance to disinfection. Treated and disinfected water should be protected from bird faeces during storage.
- *Escherichia coli* (or alternatively, thermotolerant coliforms) is a suitable indicator for the presence or absence of *Campylobacter* in drinking-water.

1 General description

Bacteria in the genus *Campylobacter* belong to the family Campylobacteriaceae. *Campylobacter* are Gram-negative, curved spiral rods (0.2 µm to 0.8 µm wide, 0.5 µm to 5 µm long) with either a single unsheathed polar flagellum, bipolar flagella, or no flagellum, depending on the species (Man, 2011; Pitkänen & Hänninen, 2017). Most species are microaerophilic (require decreased oxygen) and capnophilic (require increased carbon dioxide), but some species prefer anaerobic conditions for growth (Man, 2011; Kaakoush et al., 2015).

As of 2019, 32 species and 9 subspecies are classified in the genus *Campylobacter* (Costa & Iraola, 2019). *Campylobacter jejuni* is the most frequently isolated species from patients with *Campylobacter* gastroenteritis, accounting for 90% to 95% of infections (Pitkänen & Hänninen, 2017). *Campylobacter jejuni* has 2 subspecies, *C. jejuni* subsp. *jejuni* and *C. jejuni* subsp. *doylei*. Less frequently, *C. coli* is reported as the pathogenic agent. *Campylobacter jejuni* and *C. coli* belong to the group of thermophilic *Campylobacter* (optimum growth around 42 °C). A growing number of emerging species, including *C. lari*, *C. fetus*, *C. concisus*, *C. upsaliensis* and *C. ureolyticus*, have also been associated with gastrointestinal disease (Man, 2011). The incidence and pathogenicity of these species needs further assessment (Platts-Mills & Kozek, 2014).

2 Human health effects

Campylobacter gastroenteritis, also known as campylobacteriosis, may be asymptomatic or cause mild to severe illness. Clinical symptoms of infection are characterized by abdominal pain, acute watery diarrhoea (with or without blood or faecal leukocytes), vomiting, chills and fever. The infection is usually self-limiting, and symptoms may persist from a few days to around 2 weeks (average of 6 days) (Heymann, 2015; Kaakoush et al., 2015). Estimates of case fatality ratios range from less than 0.01% to 8.8%, depending on the population studied (Werber et al., 2013; O'Reilly et al., 2012).

In low-income countries, *Campylobacter* infection in children has been shown to be associated with malnutrition, enteric dysfunction and impaired growth (Platts-Mills & Kosek, 2014; Amour et al., 2016). The prevalence of asymptomatic infection in low-income settings is known to be high, but the consequences of this are not sufficiently understood.

Acute *Campylobacter* infections can have infrequent but serious post-infection disease syndromes in humans, including reactive arthritis (1% to 5% of infections), meningitis (rare) and Guillain-Barré syndrome (estimated to be 0.004% of infections in high-income settings) (Kuroki et al., 1991; Butzler, 2004; Pitkänen & Hänninen, 2017). Globally, approximately one third of Guillain-Barré syndrome cases have been attributed to preceding *Campylobacter* infection. There is also increasing evidence associating campylobacteriosis with inflammatory bowel disease and irritable bowel syndrome (Gradel et al., 2009; WHO, 2013).

In severe cases, treatment of campylobacteriosis with antibiotics may be beneficial. However, multiple resistant strains have been reported (Szczepanska et al., 2017), and increasing incidence of fluoroquinolone resistance in *Campylobacter* has rendered macrolides such as

erythromycin and azithromycin the drugs of choice for human campylobacteriosis (Bolinger & Kathariou, 2017). Macrolide resistance in *C. jejuni* remains low but can be common in *C. coli*.

3 Epidemiology

Campylobacter are one of the most important causes of diarrhoeal diseases worldwide (Pires et al., 2015; GBD, 2020). The epidemiology and burden of infection varies between high- and low-income settings.

In low-income settings, *Campylobacter* infection is often endemic, and illness is largely confined to children under 2 years of age (Havelaar et al., 2009; Platts-Mills et al., 2015; Liu et al., 2016). Infections in other age groups are often asymptomatic, probably due to acquired immunity. Available evidence indicates that the incidence and mortality rates from campylobacteriosis are higher in low-income settings (Pires et al., 2015). However, the burden of infection is poorly quantified in those settings as a result of weak surveillance systems, inadequate facilities for testing and high levels of asymptomatic infection (Havelaar et al., 2009; Platts-Mills & Kosek, 2014).

In high-income settings, surveillance shows that the number of reported cases of campylobacteriosis has increased in the last 10 to 15 years (Pitkänen & Hänninen, 2017). All ages can be affected, but the burden is highest in children under 5 years of age. Campylobacteriosis is a major cause of diarrhoea among travellers from high-income settings to elevated-risk settings (Domingues et al., 2012; Kaakoush et al., 2015).

In all settings, immunocompromised individuals are more vulnerable to infection, relapse and gastrointestinal and extra intestinal complications (Kaakoush et al., 2015). Infection rates display a strong seasonal variation in temperate climates (less distinct in the tropics), with infections being more common in summer than in winter (Schönberg-Norio et al., 2004; Pitkänen & Hänninen, 2017).

4 Transmission

Campylobacter are transmitted primarily by the faecal–oral route with poultry, cattle and other domestic animals, wildfowl, pets and humans all being reservoirs of infectious organisms (Mughini Gras et al., 2013; Kaakoush et al., 2015; Pitkänen & Hänninen, 2017). Food is the most common source of exposure to *Campylobacter*, with undercooked meat and meat products, as well as raw or contaminated milk, identified as leading sources of infection. (FAO/WHO, 2009; Hald et al., 2016; WHO, 2020). Direct contact with poultry, livestock and pets and consumption of contaminated drinking-water are also recognized as significant sources of exposure (Mughini Gras et al., 2013; Kaakoush et al., 2015; Pitkänen & Hänninen, 2017). Direct person-to-person transmission causes less than 5% of cases (Friedman et al., 2004; Gilpin et al., 2013; Kaakoush et al., 2015). Infrequent reports of recreational contact leading to disease have been reported (Schönberg-Norio et al., 2004).

Estimates of the infectivity of *C. jejuni* vary depending on the strain and age of those exposed, but the relative infectivity may be considered to be moderate to high¹. An ID₅₀ (50% infective dose i.e. the number of organisms required to infect 50% of exposed subjects) of 890 organisms has been estimated from a challenge study in adults with strain A3249 (using the dose response of Medema et al., 1996) but ID₅₀s below 10 organisms have been estimated for other strains and from analysis of outbreaks in children (Teunis et al., 2018). There is some evidence, particularly in adults, that infection could occur at low doses while high doses are required for illness (Teunis et al., 2018). Acquired immunity could be a contributing factor (Teunis et al., 2005). The incubation period is usually 2 to 5 days, with a range of 1 to 8 days; probably depending on dose (Percival, 2014; Pitkänen & Hänninen 2017). Gastrointestinal symptoms last on average for 6 days (Kaakoush et al., 2015), while shedding of the pathogen (6×10^6 to 10^9 colony forming units (CFU)/g of faeces) can persist for a few hours to 3 weeks (average 15 days) after infection (Pitkänen & Hänninen, 2017).

The great majority of reported cases of *Campylobacter* infections are sporadic, but common source outbreaks can occur, usually arising from contaminated food or drinking-water supplies (Kaakoush et al., 2015).

5 Detection methods

Standard culture-based methods for *Campylobacter* in water are available (e.g. ISO 17995, 2005), but they are complex compared to detection methods for other microorganisms such as *E. coli*, incorporating multiple steps and requiring specific conditions and media, including microaerobic incubation. Incubation temperatures can also influence the detection of thermophilic *Campylobacter* in wastewater and surface water (Khan et al., 2013). Molecular methods are also available, but care needs to be taken in selecting primers and validating the specificity of the assays used (Banting et al., 2016). Distinguishing between viable and non-viable cells is a shortcoming of molecular methods.

¹ From experiments with human volunteers, from epidemiological evidence and from experimental animal studies. Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: High means infective doses can be $1-10^2$ organisms or particles, Moderate 10^2-10^4 and Low $> 10^4$.

6 Source and occurrence

Various environmental matrices have been examined for *Campylobacter*. A summary of their sources, concentrations and persistence is provided in Table 1 and the subsequent text.

Table 1. Summary of *Campylobacter* occurrence data (adapted from Pitkänen & Hänninen, 2017)

Source	Prevalence		Concentration/L ^a				Method
	Studies (samples)	% positive	Studies (samples)	Averages ^b	Studies (samples)	Range	
Raw sewage	3 (169)	36–86	4 (102)	2.5×10^3 – 1.1×10^6	2 (67)	$<4.0 \times 10^1$ – 1.0×10^7	Culture
	2(69)	26, 83	1 (57)	4.1×10^6	1 (57)	$<1.0 \times 10^2$ – 6.2×10^7	PCR ^c
Wastewater sludge	3 (158)	9–90	1 (30)	2.5×10^3	1 (30)	3.2×10^2 – 6.3×10^5	Culture
Treated wastewater	3 (171)	25–53	4 (104)	3.0×10^2 – 1.3×10^3	1 (37)	$<1.0 \times 10^3$ – 1.0×10^5	Culture
	1 (12)	42	NR ^d	NR	NR	NR	PCR
Surface water	16 (2763)	2.7–86	7 (1115)	1.8×10^0 – 4.6×10^4	7 (658)	$<1.0 \times 10^0$ – 1.0×10^6	Culture
	2 (426)	31, 70	2 (426)	5.5×10^1 , 3.3×10^5	2 (426)	$<1.0 \times 10^2$ – 4.6×10^6	PCR
Groundwater	2 (153)	12, 75	1 (18)	1.2	2 (153)	<0.6 – 7.2	PCR

^a Concentrations represent cells/L by culture or calculated from standard curves using quantitative PCR.

^b Reported as averages, medians and geometric means. Values are presented as a range where more than one study has been identified.

^c Polymerase chain reaction

^d Not reported

6.1 Faecal pollution sources

In studies of raw sewage *Campylobacter* were detected by culture in 36% to 86% of samples, in concentrations up to 1×10^7 organisms/L (Table 1). Higher concentrations were detected using molecular methods. *Campylobacter jejuni* and *C. coli* were detected by culture in 23% of samples of primary sludge with mean concentrations of 278 organisms and 1403 organisms/g dry matter (Stampi et al., 1999). *Campylobacter* were detected by culture in 9% to 90% of sewage sludge, in concentrations up to 6.3×10^5 organisms/L (Table 1).

Treated wastewater generally contains lower numbers of *Campylobacter* (Pitkänen & Hänninen, 2017). After activated sludge treatment 25% to 53% of samples contained culturable *Campylobacter*. Treated, but undisinfected, wastewater contained on average between 3.0×10^2 and 1.3×10^3 organisms/L with a maximum of 1×10^5 organisms/L (Table 1).

Campylobacter were detected by culture in 36% of samples from goat, cow, chicken, goose, duck and pig faeces, in 64% of poultry faecal samples and in 24% of wildfowl faeces (6.6×10^4 to 6.6×10^6 organisms/g) (FSANZ, 2010; Jokinen et al., 2010; Ahmed et al., 2012).

Temperature and the presence of oxygen are key factors in determining inactivation rates in all environments (Pitkänen & Hänninen, 2017). A study on the survival of *Campylobacter* in manure slurry under anaerobic conditions at temperatures between 4 °C and 35 °C showed little reduction in culturable organisms (initial concentrations 1×10^7 to 1×10^8 organisms/L)

over 200 days (Kearney et al., 1993). Molecular studies showed pathogen persistence of up to 10 months in composted cattle faeces (Inglis et al., 2010). In cow pats on pasture, the time taken for a 1 log₁₀ reduction of culturable *Campylobacter* (initial concentrations 6×10^6 to 2.7×10^7 organisms/g) was 16 days in winter and 1.2 days in summer (Sinton et al., 2007). *Campylobacter* survived in livestock manure slurries for 32 days under field conditions but died in 2 to 4 days in manure heaps reaching 55 °C (Nicholson et al., 2005).

6.2 Surface water

Campylobacter are frequently present in surface waters, with various surveys detecting culturable organisms in 2.7% to 86% of samples (Table 1). Concentrations were typically in the range of 10¹ to 10⁴ organisms/L but occasionally up to 10⁶ organisms/L. The occurrence of *Campylobacter* in surface waters is dependent on rainfall, water temperature and the presence of poultry, waterfowl or other contamination sources (Pitkänen, 2013).

Campylobacter have short persistence in surface waters². Survival rates are influenced by key factors of temperature, oxygen, sunlight, interactions with other organisms, and concentrations of nutrients (Thomas et al., 1999; Pitkänen and Hänninen, 2017). *Campylobacter* can be harboured within some amoeba and persist longer (Axelsson-Olsson et al., 2005; Viera et al., 2015). In laboratory conditions, *Campylobacter* can survive at 4 °C for extended periods, while inactivation is more rapid at elevated temperatures. Using culture-based methods, 4 log₁₀ inactivation of *C. jejuni* took 30 to 90 days at 4 °C, but this was reduced to 4 to 8 days at 20 °C (initial concentrations 4.45 to 5.63 log₁₀ organisms/mL); 1 log₁₀ inactivation took 12 to 18 days at 4 °C and 1 to 4 days at 20 °C (Gonzalez & Hänninen, 2012).

6.3 Groundwater

Campylobacter can be present in vulnerable groundwater, i.e. contaminated by surface water or sewage, but only a few studies have quantified the detection rates or concentrations present (Table 1). In two studies of shallow groundwater, *Campylobacter* were detected in 12% to 75% of samples at concentrations up to 7.2 genome copies/L (Savill et al., 2001; Close et al., 2008).

6.4 Drinking-water

Detection of *Campylobacter* in adequately treated drinking-water supplies is rare. *Campylobacter* have been isolated from tap water in prospective studies of unchlorinated drinking-water supplies sourced from a vulnerable spring fed stream (Kay et al., 2007) and during the investigations of outbreaks caused by inadequately chlorinated or unchlorinated groundwater (O'Connor, 2002; Clark et al., 2003; Kuusi et al., 2004). Refer to the *Significance in drinking-water* section of this background document for further information.

² Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: Detection period for infective stage in water at 20 °C: Short, up to 1 week; Moderate, 1 week to 1 month; Long, over 1 month.

6.5 Soil and crops

There are few studies of *Campylobacter* in soil or crops. In temperate climates, *Campylobacter* in animal manure spread on fields can persist for approximately one month (Pitkänen & Hänninen, 2017).

6.6 Surfaces

Campylobacter can survive for more than 4 hours on surfaces such as stainless steel, formica and ceramic tiles (De Cesare et al., 2003). Lag times and inactivation rates were influenced by suspending material and the nature of the surface. Starting with inocula of 10^7 to 10^8 culturable organisms/mL, lag times before inactivation ranged from 65 to 190 minutes and the time for 3 log₁₀ inactivation ranged from 100 to 250 minutes.

7 Significance of sanitation

Safe sanitation systems are important barriers against the spread of *Campylobacter* (Pitkänen & Hänninen, 2017). Sewage has been identified as a source of a number of drinking-water outbreaks (Pitkänen, 2013) and septic tank waste has also been implicated as a source of an outbreak (Bopp et al., 2003). However, since animals are a major reservoir for *Campylobacter*, a sanitation service chain that separates human faeces from the environment and from human contact will only partially interrupt transmission. A comprehensive approach that also deals with the management of animal faeces is necessary to control the transmission of *Campylobacter*-related diseases (WHO, 2013).

8 Significance in drinking-water

Campylobacter are major zoonotic waterborne pathogens. There is strong evidence linking transmission of *Campylobacter* with unsafe drinking-water. In the United States of America, between 1971 and 2008, there were 19 reported outbreaks caused by *Campylobacter*, generating a total of 5608 reported cases (Craun, 2012). A review of international *Campylobacter* outbreaks between 1978 and 2009 found a total of 28 reported drinking-water associated outbreaks, generating 103 000 cases (Pitkänen, 2013). Most of the reported cases were linked to relatively small groundwater-based systems contaminated by animal faeces or human sewage, either after a heavy rain or due to disinfection failure. In some cases, waterborne transmission was confirmed by the isolation of the same strains from infected patients and the drinking-water source (Pitkänen et al., 2008). In 2000, a waterborne outbreak caused by *C. jejuni* (and *E. coli* O157:H7) occurred in the farming community of Walkerton in Ontario, Canada. The outbreak led to 7 deaths and more than 2300 illnesses. The inadequately chlorinated drinking-water supply was contaminated by rainwater runoff containing cattle faeces from neighbouring farms (O'Connor, 2002; Clark et al., 2003). Another waterborne outbreak of campylobacteriosis involving non-chlorinated groundwater occurred in 2000 in a rural municipality in Finland (Kuusi et al., 2004). The outbreak was associated with at least 463 cases and was probably caused by infiltration of faecally contaminated surface water into the groundwater supply during heavy rains.

9 Prevention and management

Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains, are needed to manage microbial risks, including from *Campylobacter*.

The effectiveness of sanitation and drinking-water treatment systems depends on the selection and operation of technologies and the characteristics of the target organisms. Treatment processes based on size exclusion (e.g. filtration through media or soil in septic tank drainage areas) or settling in primary wastewater treatment processes will be limited for removing bacteria, including *Campylobacter*. Oxidizing chemical disinfectants, such as chlorine and UV light, are effective at inactivating *Campylobacter* and other non-spore forming bacteria.

9.1 Sanitation systems

Sanitation safety plans (WHO, 2022) should include control measures to safely collect, treat and discharge or reuse wastewater and faecal sludges. Removal or inactivation of *Campylobacter* by wastewater treatment processes is variable, although the data are limited (Pitkänen & Hänninen, 2017). Based on available evidence, the reduction of *Campylobacter* by wastewater treatment processes is likely to be similar to the reduction of faecal indicator bacteria such as *E. coli* (Pitkänen & Hänninen, 2017).

Little is known about *Campylobacter* reduction by onsite treatment of human faeces, but it is assumed to be similar to that for animal manures. In experiments with swine manure, *C. coli* remained viable for up to 24 days at 4 °C and 1 week at 15 °C but only a few hours at 42 °C (Bui et al., 2011).

In centralized sewage treatment plants, bacteria such as *Campylobacter* are not very sensitive to removal by size exclusion or settling processes, and it is expected that clarifiers or sedimentation tanks will generally achieve less than 1 log₁₀ reduction in concentrations (Oakley, 2018). The concentrations of *Campylobacter* in raw sewage can be as high as 10⁷ CFU/L, and treated wastewater has been reported to contain up to 10⁵ CFU/L (Pitkänen & Hänninen, 2017). Typical reductions of *Campylobacter* are less than 2 log₁₀ for secondary processes (e.g. activated sludge, stabilization ponds) (Table 1) (Pitkänen & Hänninen, 2017) provided turbidity is low (Pitkänen & Hänninen, 2017).

Because potentially viable *Campylobacter* may be present in sewage sludge, faecal sludge, and septage, special attention must be given to solids management at onsite and centralized facilities. Not all sludge treatment processes are equally effective at inactivating *Campylobacter*. For example, mesophilic anaerobic digestion (35 °C) is relatively ineffective for *Campylobacter*, as they typically survive better than faecal indicator bacteria (Whiley et al., 2013). Inactivation is often negligible and usually less than 1 log₁₀ (Kearney et al., 1993; Horan et al., 2004). *Campylobacter* are, however, more susceptible to aerobic treatment processes and high temperatures (Betaie & Jones, 1990; Stelzer & Jacob, 1991; Pitkänen & Hänninen, 2017). The storage of dewatered sludge biosolids at 50 °C resulted in more than 4 log₁₀ reduction of culturable *C. jejuni* within a day (initial concentrations of 1.8 × 10³ to 1 × 10⁵ CFU/g) (Ahmed & Sorensen, 1995). The same reduction took 6 days at 38 °C and 11 days at

22 °C, while at 5 °C, only a 2 log₁₀ reduction was observed within a 62-day period (Ahmed & Sorensen, 1995).

9.2 Drinking-water supplies

Within a water safety plan (WHO, 2012; 2023), control measures to manage potential risks from *Campylobacter* should include protection of source waters (surface or groundwater) from contamination by animal and human faecal wastes (including extra vigilance during heavy rains), application of adequate water treatment processes and protection of drinking-water quality during distribution and storage. Treated and disinfected water should be protected from bird faeces during storage. *Campylobacter* have low resistance³ to chlorine and monochloramine with 2 log₁₀ inactivation achieved at a Ct (product of disinfectant concentration and contact time) value of less than 0.5 mg.min/L for free chlorine and less than 5 mg.min/L for monochloramine at pH 6 to 8 and 4 °C to 25 °C (Blaser et al., 1986). *Campylobacter* are also sensitive to UV light; a dose of 7 mJ/cm² can provide a 2 log₁₀ reduction (Hijnen et al., 2006).

Escherichia coli (or alternatively, thermotolerant coliforms) is a suitable indicator for the possible presence or absence of *Campylobacter* in drinking-water.

10 ICD-11 classification

XN7US	<i>Campylobacter</i>
XN0BA	<i>Campylobacter coli</i>
XN3EN	<i>Campylobacter fetus</i>
XN4Q5	<i>Campylobacter jejuni</i>
1A06	Gastroenteritis due to <i>Campylobacter</i>
1C40	Campylobacteriosis
MG50.1	Antibiotic resistant <i>Campylobacter</i>

³ Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: Resistance is based on 99% inactivation at 20 °C where, generally, Low represents a Ct₉₉ of < 1 min.mg/L, Moderate 1–30 min.mg/L and High > 30 min.mg/L (where C = the concentration of free chlorine in mg/L and t = contact time in minutes) under the following conditions: the infective stage is freely suspended in water treated at conventional doses and contact times, and the pH is between 7 and 8. It should be noted that within pathogen species and groups, there are likely to be variations in resistance, which could be further impacted by characteristics of the water supply and operating conditions. Organisms that survive and grow in biofilms will be protected from chlorination.

Selected references

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Annex 1. Approach to content development and declarations of interest

The background documents were developed drawing from the following resources (i) microbial fact sheets from the fourth edition of the WHO Guidelines for Drinking-Water Quality (ii) pathogen-specific chapters in the Global Water Pathogen Project (GWPP) book “*Sanitation and Disease in the 21st Century*”, an update of the seminal book by Feachem et al. 1983 “*Sanitation and Disease: Health Aspects of Excreta and Wastewater Management*” and (iii) a rapid review of the literature at the start of the development of the background documents and before finalization in December 2023, focusing on identifying additional recent high quality literature reviews on a given pathogen. The documents were prepared by members of the WHO drinking-water quality microbiology group as well as external scientists with expertise in the relevant pathogens. The documents went through several iterations based on inputs from members of the WHO drinking-water quality microbiology working group and WHO technical staff. In addition, each document was reviewed by at least two external peer reviewers including authors of the GWPP book chapters. The extensive review process was designed to ensure technical accuracy and to identify any data gaps. The background documents were discussed at three expert meetings to resolve any issues and ensure the content reflected a consensus of expert opinions. Upon satisfactory resolution of any further data gaps or issues, the background documents went through technical editing before finalization.

Before the key expert meetings mentioned above, and for this work more broadly, all microbiology working group members submitted declarations of interest to WHO. These were to disclose any potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence in relation to the subject matter of the guidance. WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects discussed at the meetings or covered by the guidance.